iRpgFormatter for RDi

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Overview

The iRpgFormatter formats the DCL-* statements of RPGLE or SQLRPGLE sources:

- Execute from Remote System filters.
- Execute from Project Explorer items.
- Execute from Lpex view.
- Execute on save.
- Optionally executes the IBM formatter (only in Lpex editor).

iRpgFormatter is an open source plug-in for RDi 9.6.0.13+. It formats free-form RPG IV source code to improve readability and consistency.

Feature list:

- Formats free-form RPG IV source code.
- Various formatting options:
  - Casing style.
  - Add/remove const() keyword.
  - Break name on case change.
  - Sort const/value to end.
  - Replace dcl-pi name with *N.
  - Remove end-proc name.
  - Unindent compiler directives.
- Optionally executes the IBM formatter.
- Optionally format source on save.

iRpgFormatter is an Open Source project that is hosted on github.com.

Limitations

The formatter does not have an Abstract Syntax Tree (AST), which usually is required for formatting source code. But it has basic knowledge about functions, keywords, special words and data types.

Most of the time it produces the expected result. But there are edge cases where the result is unexpected. Unexpected results can occur if the structure of the source code is not completely visible in the source member, but hidden in includes or manipulated by conditional compiling.

For example:

```
/if defined(DEFINE_PROCPTR)
dcl-pr SrvPgmLayout_new ind extproc(g_pLayout_new);
/else
dcl-pr SrvPgmLayout_new ind extproc('SrvPgmLayout_new');
/endif
  i_pSuper   like(LOG4RPG_pLayout_t) const;
  io_pHandle pointer;
  i_propList like(LOG4RPG_pPropertyList_t) const;
  i_prefix   like(LOG4RPG_prefix_t) const;
end-pr;
```

The problem of the example above is that there are two `dcl-pr` statements in a row. This edge case should work, because the block statements are automatically closed if a block statement is followed by the same type.

Menu Options

The *iFormat* menu option is in the *Source* sub-menu. You can map a user key action to it on the **LPEX Editor** – **User Key Actions** page.

Preferences

The preferences page has been placed next to the IBM *Formatter* preferences page and is labeled as *iRpgFormatter*.

Play with the settings and watch the preview panel to understand the formatting options.

The following tabs contain lists of entries used for case-sensitive formatting:

- Data types
- Declaration types
- Keywords
- Special Words

A more detailed description of the settings can be found on the iRpgFormatter project page on GitHub.

Option "Execute IBM formatter" only works if the formatter is started in the Lpex editor.

Change Log 1.0.7.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.
- Fix incorrectly committed source code.

Change Log 1.0.6.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.

Change Log 1.0.5.r (01.06.2026)

Plug-in: 9.6.0.13+

- Fix `UnexpectedTokenException` when formatting keywords.

Change Log 1.0.4.r (25.05.2026)

Plug-in: 9.6.0.13+

- Fix `ClassNotFoundException` in RDi 9.6.
- Applied experimental edge case fix: compiler conditions.
- Fix formatting compiler directives.
- Allow spaces between function name and parameter list.
- Fix setting maximum line length.

Change Log 1.0.3.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-pi *n end-pi;` parser error.
- Fix `Line n not found.` error in Lpex editor when positioning the cursor at the end of a formatting action.

Change Log 1.0.2.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Unexpected token of type: FUNCTION` exception.

Change Log 1.0.1.r (12.05.2026)

Plug-in: 9.6.0.13+

- Add formatting of compile time arrays.

Change Log 1.0.0.r (12.05.2026)

Plug-in: 9.6.0.13+

- Initial version

iRpgFormatter On The Internet

Project page on GitHub	https://github.com/tools-400/irpgformatter-plugin/
Homepage	https://tools-400.github.io/irpgformatter/
RDi 9.6+ Update Site	https://tools-400.github.io/irpgformatter/update-site/eclipse/rdi8.0/
Eclipse Marketplace	https://marketplace.eclipse.org/content/irpgformatter-rdi-96013/
RDi 9.6 Beta Update Site	https://tools-400.github.io/irpgformatter/beta-version/update-site/eclipse/rdi8.0/